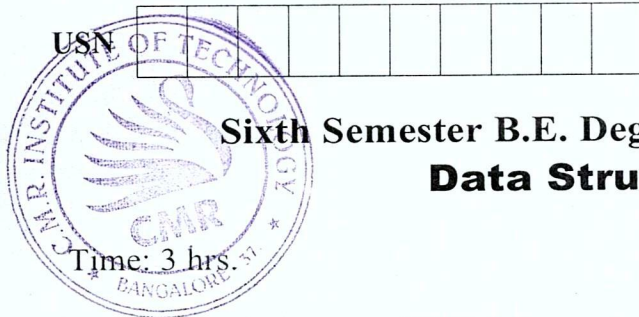




Sixth Semester B.E. Degree Examination, June/July 2025
Data Structures using C++

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define C++ and explain its features. (06 Marks)
- b. With example, explain the different expressions used in C++. (07 Marks)
- c. What is constructor? Discuss the different types of constructors with example. (07 Marks)

OR

- 2 a. Explain operator overloading with C++ program. (08 Marks)
- b. Explain the different types of Inheritance. (06 Marks)
- c. Explain Recursion. Write a recursive function in C++ to find the factorial of a number. (06 Marks)

Module-2

- 3 a. Define sparse matrix and also explain the representation of sparse matrix using single linear. (08 Marks)
- b. Write the abstract class linear list. (06 Marks)
- c. Write a program in C++ to perform create, display operation on single linked list. (06 Marks)

OR

- 4 a. What is a linear list? write the abstract class of linked list. (06 Marks)
- b. How dynamic memory allocation is performed in C++? Explain with suitable example. (08 Marks)
- c. Write struct definition for chain node. (06 Marks)

Module-3

- 5 a. Write C++ abstract class for stack. (06 Marks)
- b. Using stack change the following infix to postfix expressions:
 $(A + B) * (C - D) * (F + C)$ (06 Marks)
- c. Explain the concept of towers of Hanoi problem using stack. (08 Marks)

OR

- 6 a. Explain how parenthesis matching is carried out using C++. Write C++ program (function) for the same. (10 Marks)
- b. Explain the evaluation of postfix expression using stack with example. (10 Marks)

Module-4

- 7 a. Write ADT specification of a Queue. (06 Marks)
- b. What is Hashing? Explain hashing function and tables. (06 Marks)
- c. Write ADT specification and abstract class for dictionary. (08 Marks)

OR

- 8 a. Discuss the problem description and solution strategy for rail road car management. (10 Marks)
- b. Explain the various operations to be performed on dictionaries with examples. (10 Marks)

Module-5

- 9 a. What is Binary Tree? Explain the properties of binary trees in detail. (10 Marks)
- b. Write a binary tree for the algebraic expression :

$$a + (b - c) * \left[\frac{(d - e)}{(f + g - h)} \right]$$
 (06 Marks)
- c. Explain binary search tree with simple example. (04 Marks)

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OR

- 10 a. Explain the various traversal methods of binary tree. (06 Marks)
- b. Write a C++ function to determine the height of the tree. (06 Marks)
- c. What is max heap? Write a program to initialize a max heap. (08 Marks)

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